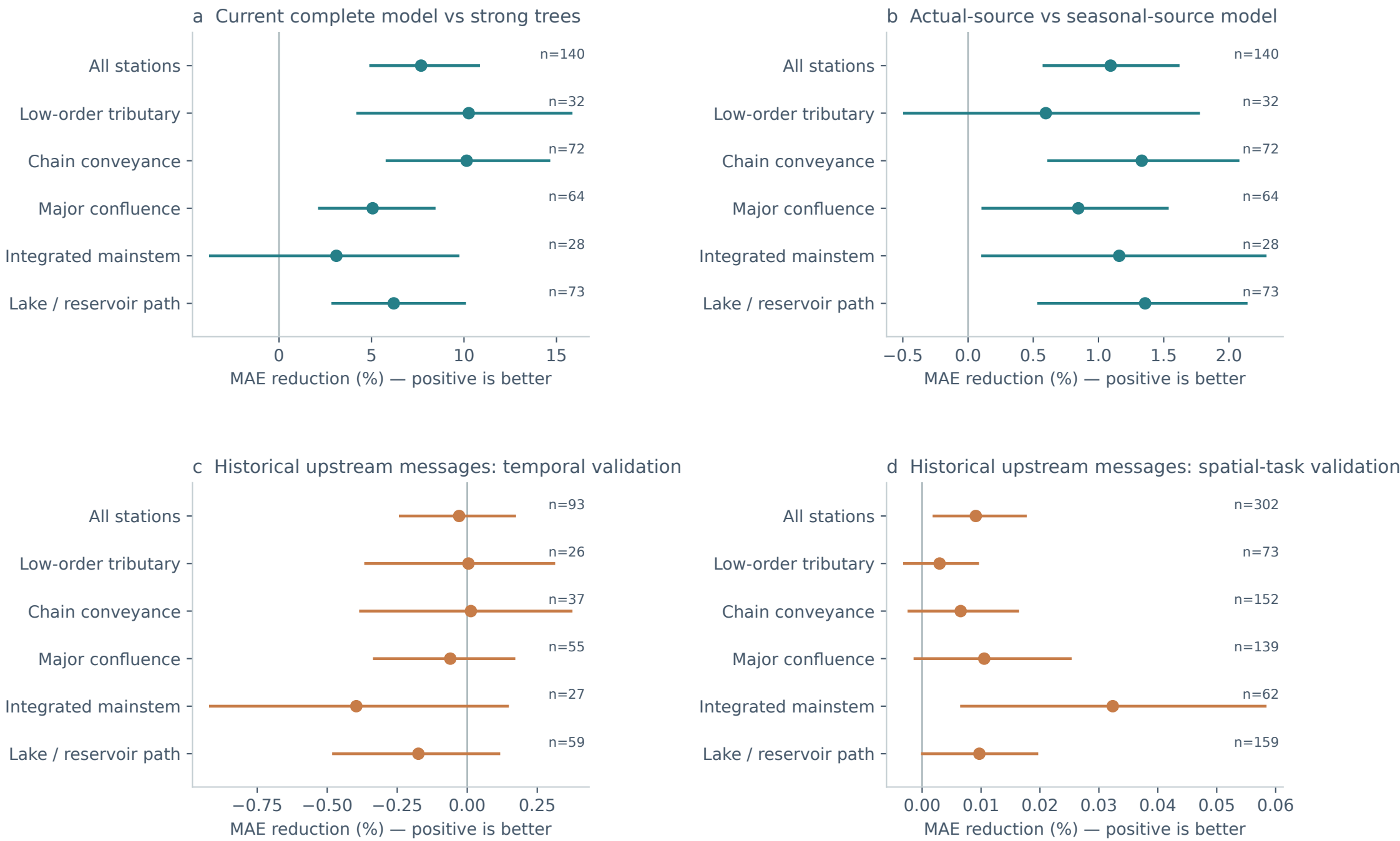


Where model information adds value



Points and 95% paired station-bootstrap intervals (5,000 draws); n denotes distinct stations. Losses are averaged across training seeds.
Top: selected source-validation roles, no physical river messages. Bottom: separate older validation tasks, upstream vs no-message residual.
Historical spatial-task validation includes source stations and cannot establish new-station transfer. Panels use independent x scales.